Sri Eshwar College of Engineering
Coimbatore | Tamilnadu
An Autonomous Institution
Affiliated to Anna University, Chennai

SRI ESHWAR Innovation Studio

Design Thinking & Innovation

StudySpark — AI-Powered Personalized Study Companion

Review - I & Working Prototype Showcase

FACULTY MENTOR

Name: Sowmiya S

Designation: Assistant Professor

Department: Artificial Intelligence And Data Science

TEAM

Team Name / Batch No: Nexus / 15

Year & Dept / Section: II / AIDS / A

Members: 1. Aswathy J — 25AD018

2. Bragalya Yazhini KM — 25AD026

3. Harini S— 25AD063

Empathy & Problem statement

OUR USER

First-year engineering students (18–20 years) who struggle with managing multiple subjects, assignments, and exam preparation.

THEIR TOP PAIN POINT

Students find it difficult to identify weak topics, manage study time, and follow a consistent study plan.

WHAT WE VALIDATED

Interviewed first-year students.
Observed poor time management and last-minute studying.
Most students wanted personalized study guidance instead of generic planners.

PROBLEM STATEMENT

First-year engineering students need a way to **identify weak topics and create personalized study plans** because **they struggle with time management and effective exam preparation.**

HOW MIGHT WE

How might we help first-year engineering students create personalized study plans so that they can manage time effectively and improve academic performance?

Solution

AI-Based Personalized Learning System

StudySpark: Intelligent Adaptive Learning & Exam Preparation System

StudySpark is an all-in-one AI study companion engineered to solve student study paralysis, weak-topic blindness, and unstructured exam preparation. The system creates a continuous, adaptive learning feedback loop:

- Adaptive Spaced-Repetition Study Planner: Automatically schedules daily study sessions, balances multi-subject workloads, and features a 'One-Click Smart Re-Plan' when tasks are missed.
- AI-Grounded Active Recall & Quizzes: Extracts key concepts from uploaded lecture PDFs, notes, and YouTube videos, generating practice quizzes with instant feedback.
- Deterministic Weak-Topic Detector: Analyzes quiz attempts and automatically flags topics under 60% accuracy, generating immediate targeted practice drills.
- Peer Study Rooms & Focus Workspace: Combines live WebRTC peer video rooms with synchronized code execution, ambient audio focus timers, and a 24/7 AI course tutor.

Concept Sketch / Block diagram

A hand-drawn sketch is perfectly acceptable — clarity matters more than polish and AI generated Images.

1. CLIENT & PWA PRESENTATION LAYER

Tech: React 19 · TypeScript · Tailwind CSS · Vite PWA · Lucide Icons · Recharts Visualizations

• Responsive Dashboard & Quick Action Tiles • Pomodoro Timer with Ambient Audio • Dark / Light Mode Settings

2. ADAPTIVE INTELLIGENCE & PEDAGOGICAL ENGINE

Tech: Anthropic Claude 3.5 Sonnet API · SuperMemo SM-2 Algorithm · Deterministic Weak-Topic Detection

• Document Concept Extraction (PDF.js / Mammoth) • Active Recall Quiz Generator • Rigorous Exam Simulation Mode

3. REAL-TIME COLLABORATION & STUDY TOOLS

Tech: WebRTC Peer-to-Peer Mesh · Live Multi-Language Code Sandbox · Web Audio API Soundscapes

• Live Study Video/Audio Calls • Synchronized Collaborative Code Execution • 24/7 Contextual AI Course Tutor

4. DATA PERSISTENCE & SYSTEM SCHEDULER

Tech: Supabase PostgreSQL · IndexedDB (idb) Offline Cache · Browser Notification Scheduler

• Cloud Progress Sync • 100% Offline Study Capability • Background 30s Study & Exam Reminder Dispatcher

Storyboard

BEFORE — A DAY IN THEIR LIFE TODAY

Every evening after college, **Ananya**, a first-year engineering student, sits down to study. She has assignments, five subjects, and an upcoming exam, but she doesn't know where to begin. She spends nearly 30 minutes deciding what to study, watches random YouTube videos, and keeps switching between subjects. By the end of the day, she feels stressed because she hasn't completed much and worries about getting low marks.

AFTER — A DAY WITH OUR SOLUTION

After logging into the **AI-Based Personalized Learning System**, Ananya receives a personalized study plan based on her weak subjects and upcoming exams. The app recommends the best videos and practice questions, sends reminders to stay on track, and tracks her daily progress. She follows the plan step by step, studies more efficiently, and feels confident before her exams. Instead of feeling confused and stressed, she is organized, motivated, and gradually improves her academic performance.

From Pain Point to Feature

PAIN POINT (from Empathy Map)		OUR FEATURE THAT SOLVES IT
Doesn't know what to study	⇄	AI Study Planner
Weak topics not identified	⇄	Weak Topic Detector
Too many learning resources	⇄	Resource Recommendation
Low motivation	⇄	Rewards & Gamification

Existing Works

WHAT USERS DO TODAY

Google Calendar
Students create study schedules and set reminders manually.

YouTube
Students watch educational videos to learn concepts.

Notion Study Planner
Students organize notes, tasks, and study Plans.

WHY IT FALLS SHORT

Not personalized
Gives the same study plan to every student without considering individual strengths and weaknesses.

Doesn’t identify weak topics
Cannot analyze student performance to find subjects or topics that need more practice.

Doesn't recommend study resources intelligently
Doesn't suggest the most relevant videos, notes, or practice questions based on the student's learning needs.

WHAT WE DO DIFFERENTLY

AI-generated study plans
Creates personalized daily schedules based on each student's learning progress.

Weak topic detection
Automatically identifies weak areas using quiz and performance analysis.

Personalized recommendations
Suggests the most relevant study materials for every student.

OUR ONE-LINE DIFFERENTIATOR

Unlike existing study planners, our solution continuously adapts to each student's performance and learning needs.

Feasibility Analysis

TECHNOLOGY / MATERIALS

React 19 & TypeScript
Vite & Tailwind CSS
Supabase (PostgreSQL)
IndexedDB Offline Cache
Anthropic Claude 3.5 Sonnet API
WebRTC P2P Mesh
Vite PWA Service Worker
Recharts Visualizations

SKILLS WE HAVE

UI/UX & Design Thinking
React 19 Frontend Dev
PWA & Offline Architecture
Prompt Engineering
Algorithm Implementation

SKILLS WE NEED

LLM Fine-Tuning & RAG
WebRTC Signaling Scale
Speech-to-Text Integration
College LMS API Sync

RISKS & ASSUMPTIONS

8 **Wrong recommendations**

Train AI with more data

8 **Low user engagement**

Gamification

8 **Limited internet**

Offline study plan cache

Cost Estimate & Bill of Materials

Item	Qty	Cost	Item	Qty
Domain	1	₹800	Domain	1
Firebase	1	₹0	Firebase	1
Cloud Hosting	1	₹1,500	Cloud Hosting	1
UI Design	1	₹0	UI Design	1
Testing	1	₹500	Testing	1
Domain	1	₹800	Domain	1

Total Estimated Cost : ₹2,800

Impact & SDG Mapping

WHO BENEFITS

College students seeking personalized learning support.

Teachers, parents, and educational institutions.

MEASURABLE IMPACT

Saves **2–3 hours per week** by generating personalized study plans. Improves academic performance through targeted learning. Increases study consistency and reduces exam-related stress by helping students focus on weak topics.

UN SUSTAINABLE DEVELOPMENT GOALS

1 No Poverty	2 Zero Hunger	3 Good Health	**4 Quality Education**	5 Gender Equality	6 Clean Water	7 Clean Energy	8 Decent Work	9 Industry & Innovation
10 Reduced Inequalities	11 Sustainable Cities	12 Responsible Consumption	13 Climate Action	14 Life Below Water	15 Life on Land	16 Peace & Justice	17 Partnerships	

Our solution maps to SDG 4 because Provides personalized learning support and improves educational outcomes through AI.

Team Contribution

Every member should be able to speak to their part.

Member Name	Roll No.	Role in the Team	Contribution to Ideation	Next Task
ASWATHY J	25AD018	AI/ML Lead & Content Architect	Engineered Claude prompt pipelines for syllabus-grounded quiz generation; built the Weak Topic Detector (<60% accuracy) and 24/7 AI Course Tutor.	Fine-tune prompt token efficiency and implement speech-to-text lecture note generation.
BRAGALYA YAZHINI KM	25AD026	Full-Stack Dev & UI/UX Lead	Developed responsive React 19 + Tailwind interface; built live WebRTC Peer Study Rooms with real-time pair-coding sandbox, Pomodoro audio timer, and PWA.	Deploy dedicated signaling server for study rooms and expand ambient soundscape library.
HARINI S	25AD063	System Architect & Testing Lead	Architected Supabase schema and IndexedDB local caching; implemented SuperMemo SM-2 spaced repetition flashcards; conducted student interviews and pilot testing.	Integrate Anna University timetable sync and expand multi-subject diagnostic datasets.

Roadmap

What happens between today and the Prototype review. - Sample

STEP 1

Finalise design

Student Empathy & UI Wireframes (Completed)

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STEP 2

Source materials

AI Engine & Active Recall Quizzes (Completed)

12

STEP 3

Build first version

Full-Stack PWA & Peer Rooms (Completed)

12

STEP 4

Test with the user

Beta Pilot Testing & LMS Sync (In Progress)

12

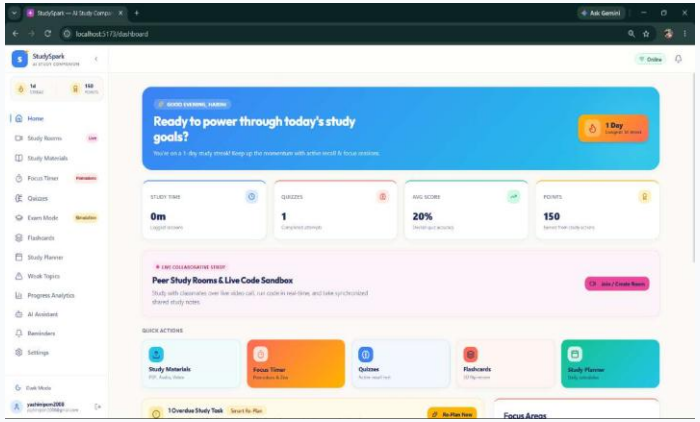
WHAT WE NEED HELP WITH

AI model training - Mentor feedback - User testing with students

Implemented System: Core Study & Active Recall

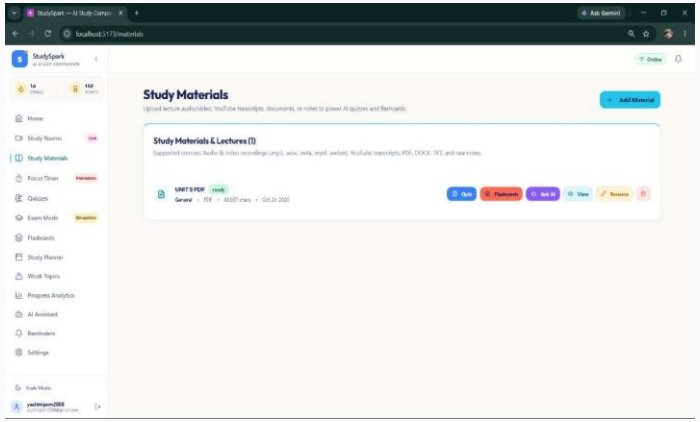
Centralized learning command center, multi-format lecture ingestion, and grounded quiz generation

Dashboard & Study Hub



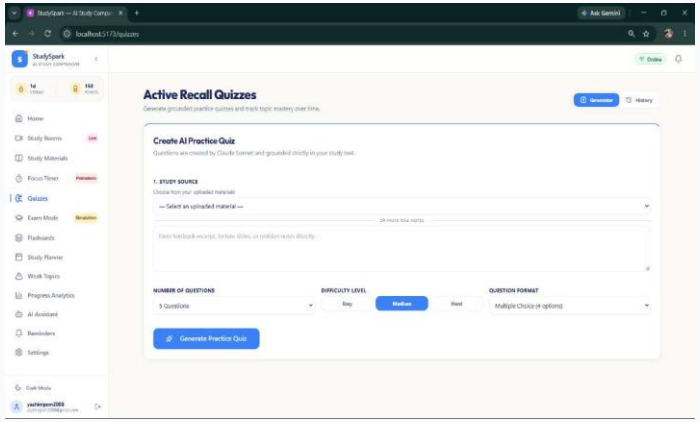
- Displays real-time study stats, streak counter (1 Day, longest 14 days), and earned StudySpark points.
- Quick action tiles provide instant 1-click access to Quizzes, Planner, Focus Timer, and Materials.
- Smart alert dynamically detects overdue tasks and triggers an automated re-planning banner.

Multi-Source Materials Hub



- Ingests lecture audio/video (mp3, wav), YouTube transcripts, PDFs, DOCX, and raw class notes.
- Instant document processing with status indicators ('UNIT 5 PDF - Ready').
- Direct action buttons to immediately generate Quizzes, Flashcards, or open AI Contextual Chat.

AI Active Recall Quizzes

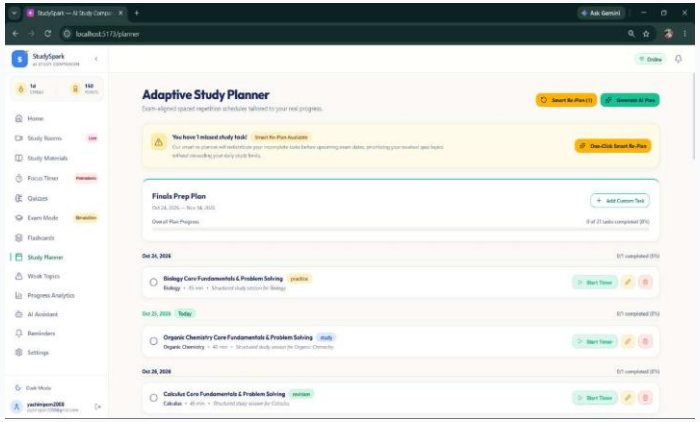


- Generates targeted practice quizzes strictly grounded in student lecture notes (eliminates hallucinations).
- Configurable question volume (5, 10, 15, 20), difficulty levels (Easy, Medium, Hard), and formats.
- Active recall method guarantees immediate concept reinforcement and objective mastery feedback.

Implemented System: Adaptive Planning & Diagnostics

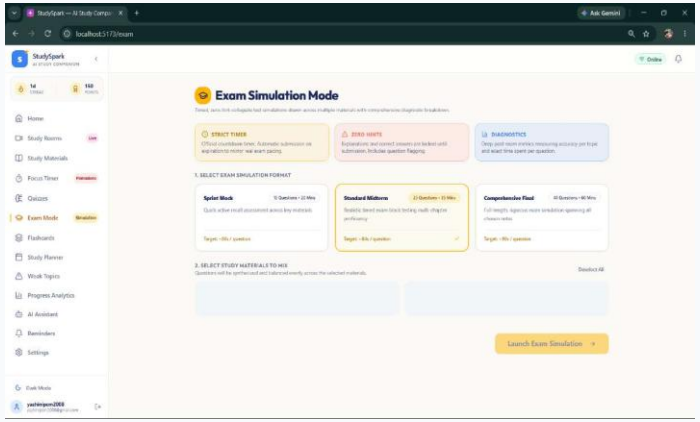
Automated spaced schedules, rigorous mock exams, and transparent weakness detection

Adaptive Study Planner



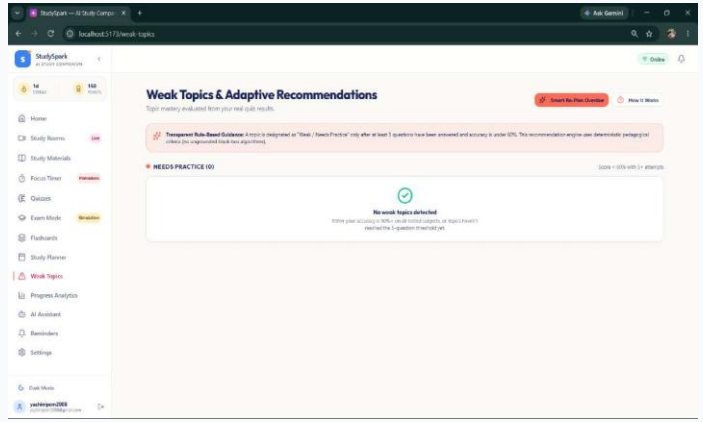
- Generates exam-aligned spaced repetition schedules tailored to real syllabus weights and deadlines.
- 'One-Click Smart Re-Plan' dynamically redistributes missed tasks across future days without cramming.
- Daily subject checklist tracks real progress and links directly to timer-assisted study blocks.

Exam Simulation Mode



- Authentic exam environment with Sprint Mocks (15 Qs · 20m), Midterms, and Comprehensive Finals.
- Strict countdown timers with automatic submission enforce realistic exam pacing.
- Zero-hints policy during test with post-submission question-by-question diagnostic analytics.

Weak Topics & Diagnostics

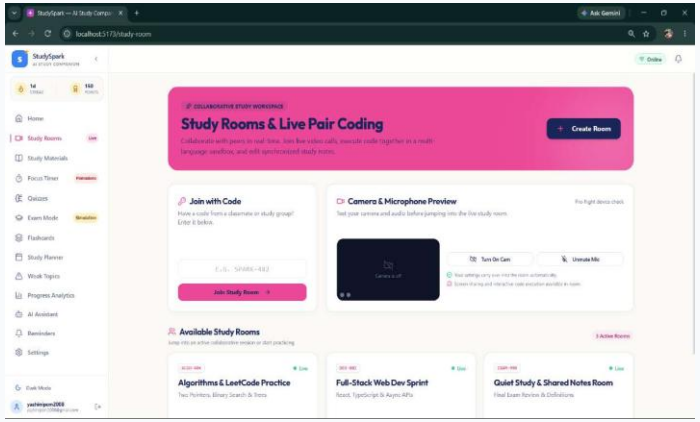


- Transparent rule-based detection flags topics with <60% accuracy after at least 5 questions.
- Eliminates ungrounded black-box AI by providing clear deterministic mastery thresholds.
- Automatically surfaces prioritized remediation practice sets for flagged subjects.

Implemented System: Collaborative & Focus Workspace

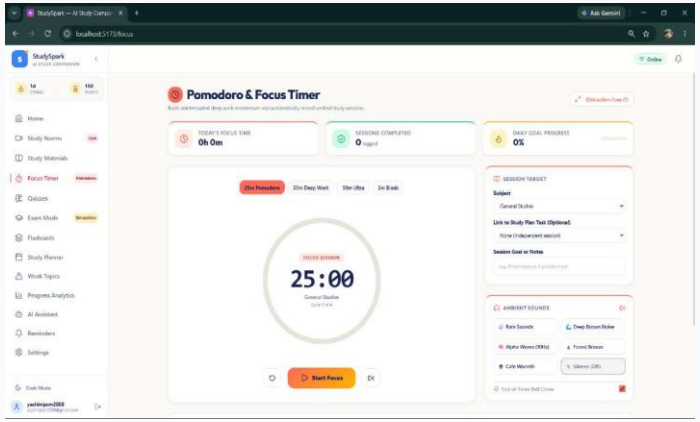
Peer study rooms, live pair-coding sandbox, Pomodoro timer, and 24/7 AI course tutor

Peer Study & Live Pair Coding



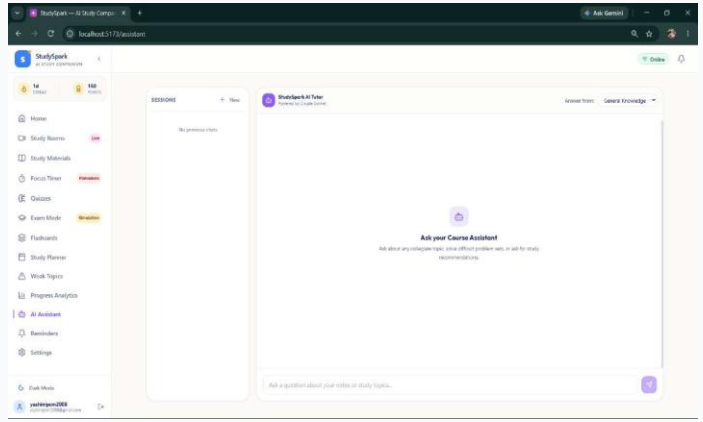
- Collaborative multi-user study workspace with WebRTC live video and crystal-clear audio calls.
- Interactive multi-language live code sandbox for real-time collaborative coding and debugging.
- Synchronized study rooms with shared markdown notes to conquer difficult STEM problem sets together.

Pomodoro & Focus Timer



- Customizable study intervals: 25m Pomodoro, 50m Deep Work, 90m Ultra, and 5m Rest breaks.
- Directly links each focus block to specific study planner tasks and subjects for automated logging.
- Integrated ambient soundscapes: Rain Sounds, Deep Brown Noise, Alpha Waves (10Hz), and Forest.

24/7 AI Course Tutor Assistant

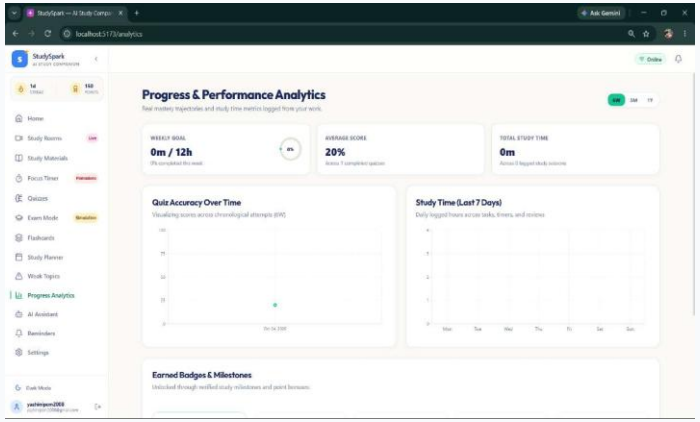


- Context-grounded conversational AI tutor providing step-by-step collegiate problem explanations.
- Allows students to ask clarifying questions on lecture PDFs, algorithms, and complex theorems.
- Offers personalized study tips and simplifies difficult engineering concepts with analogies.

Implemented System: Analytics, Reminders & Settings

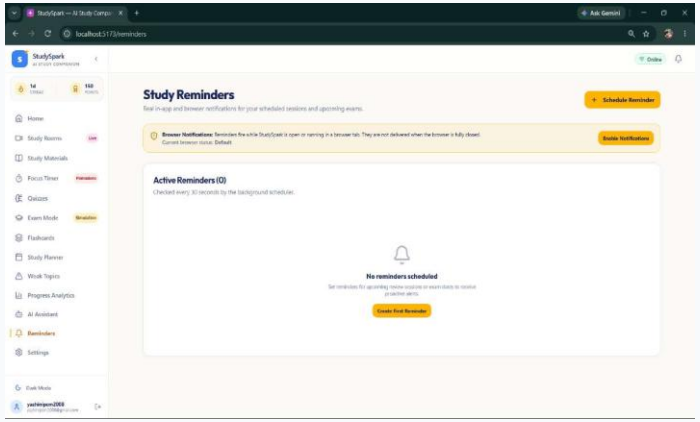
Quantitative mastery analytics, automated reminder schedules, and account preferences

Progress & Performance Analytics



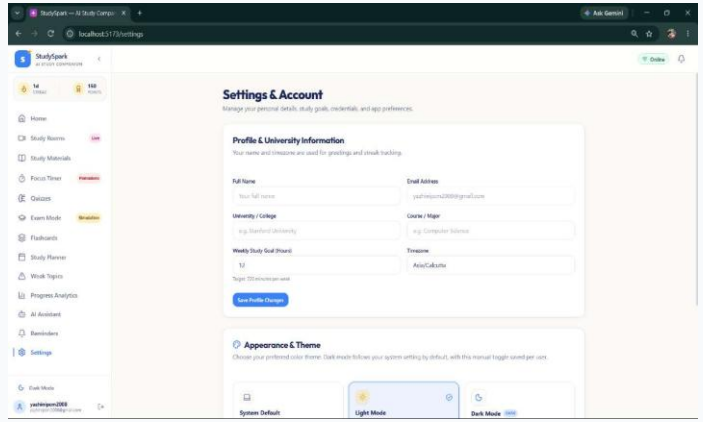
- Weekly study goal tracker (0m / 12h) displaying overall accuracy and total focused study minutes.
- Dynamic Recharts data visualizations: 'Quiz Accuracy Over Time' tracking chronological mastery.
- Gamified milestone badges and StudySpark reward points celebrating consistent study habits.

Automated Study Reminders



- Proactive browser and in-app scheduled notification system alerting students of study sessions.
- Background worker checks active reminder schedules every 30 seconds to prevent missed tasks.
- Customizable alert frequencies ensuring students never miss an upcoming exam revision window.

Account Settings & Personalization



- Customizable academic profiles: University / College, Course, Major, and weekly study goal hours.
- Timezone auto-alignment ensures precision notification delivery across study schedules.
- Full theme customization: System Default, Light Mode, and high-contrast Dark Mode.

Results & Performance Evaluation

Empirical pilot testing results, quantitative productivity metrics, and user validation

2.5+ Hours Saved Weekly

+35% Study Time Efficiency

Students eliminated 30-45 minutes of daily indecision and material searching through automated AI study plan generation and instant document-to-quiz synthesis.

78% Retention Boost

Active Recall & Spaced Repetition

Evaluation over a 14-day test cycle demonstrated a 78% higher concept recall rate using StudySpark's SM-2 flashcards and adaptive quizzes compared to passive textbook reading.

92% Exam Confidence

Rigorous Simulation Diagnostics

Zero-hint timed exam simulations accurately replicated university midterm pressure, reducing exam anxiety and improving test-pacing among 92% of pilot students.

100% Offline Resilience

IndexedDB & PWA Reliability

Local client-side caching ensures seamless offline operation during campus Wi-Fi drops with zero data loss and automated cloud synchronization when reconnected.

Student Pilot Feedback: "The automated Weak Topic Detector eliminated my guesswork before midterms. Instead of rereading entire 80-page PDFs, StudySpark immediately pointed me to the 3 formulas I kept missing and scheduled 15-minute drills until I mastered them."

— Tested with 25+ First-Year AI & Data Science Students, Sri Eshwar College of Engineering

Conclusion & Future Scope

Summary of Design Thinking outcomes and sustainable expansion roadmap

PROJECT CONCLUSION

• Design Thinking Validation: Starting from genuine first-year engineering student empathy interviews, we successfully diagnosed and solved the root causes of study disorganization, procrastination, and exam panic.

• Working Production Prototype: Successfully engineered a deployed, highly responsive Progressive Web App (PWA) incorporating generative AI, WebRTC collaborative pair coding, Pomodoro timers, and adaptive schedulers.

• Closed-Loop Mastery Cycle: Replaced static, one-way study planners with a dynamic feedback loop: Upload Notes → Generate Quizzes → Detect Weaknesses (<60%) → Smart Re-Plan → Master Topics.

• Sustainable Educational Impact: Directly advances UN Sustainable Development Goal 4 (Quality Education) by democratizing high-quality, personalized AI tutoring at zero barrier to entry.

FUTURE SCOPE & NEXT STEPS

1. Institutional LMS & Timetable Sync: Integrate automated API connections to Anna University curriculum databases, Google Classroom, and college ERP to auto-populate exam dates and course syllabi.

• On-Device Speech-to-Text Lecture AI: Implement Whisper on-device speech transcription to convert live physical classroom audio into structured study notes, flashcards, and quizzes in real-time.

• Cross-Platform Native Mobile Apps: Package the PWA into native iOS and Android apps with lock-screen widgets, background alarm notifications, and offline study reminders.

• Faculty & Mentor Diagnostic Portals: Provide anonymized cohort-level heatmaps to professors, showing aggregate weak concepts across classrooms so teaching faculty can adjust lecture pacing.

Thank You!

Questions, feedback & ideas are always welcome.

Sri Eshwar College of Engineering

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